

(Chapter 1)

Delta Lake origins to solve

→ Petabyte scale systems data consistency & integrity.

Data warehouse has the limitation of

→ hard to scale

→ data velocity adaption. [Batch or stream ingestion]

So data lakes were needed.

- ② Data Lake can handle internet ~ scale volume, velocity, variety.

Data warehouse (single node vertical scaling mostly structured data)

Data Lake $\rightarrow$ [distributed cheap commodity hardware, ~~also~~ unstructured support as well].

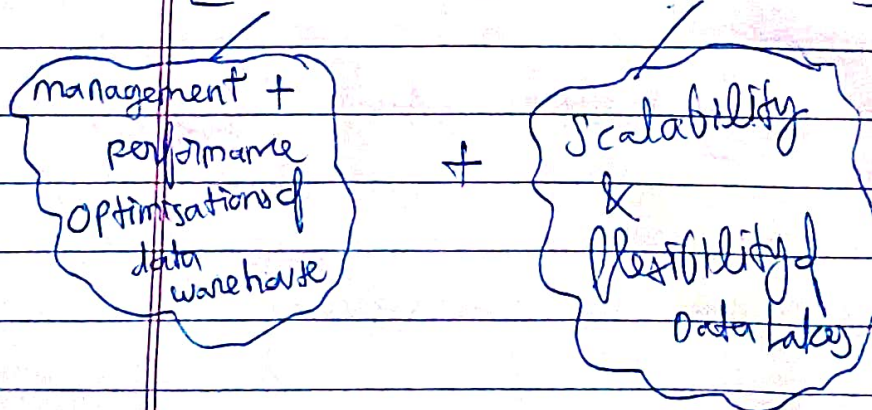
~~But~~ But don't provide ACID.
they provide

BASE $\rightarrow$ $\times$ basically available
 $\times$ soft-state
 $\times$ eventually consistent

It will be inconsistent

- ③ Lakehouse (or Data Lakehouse)

[Data Warehouse + Data Lake]



classmate

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got to know full form of power(BI)
[Business Intelligence]

Initially delta was built to work only
with Spark.

But it can work with Spark, Flink, Trino
Presto as well now.